

Education

H.B.Sc., University of Toronto

2018-2022

Astronomy & Astrophysics, Statistics, Mathematics. Dean's List Scholar.

Publications

1. **Jeff Shen**, Gwendolyn M. Eadie, Norman Murray, Dennis Zaritsky, Joshua S. Speagle, Yuan-Sen Ting, Charlie Conroy, Phillip A. Cargile, Benjamin D. Johnson, Rohan P. Naidu, Jiwon Jesse Han. The Mass of the Milky Way from the H3 Survey, submitted to ApJ.
2. **Jeff Shen**, Allison W. S. Man, Johannes Zabl, Zhi-Yu Zhang, Mikkel Stockmann, Gabriel Brammer, Katherine E. Whitaker, and Johan Richard. Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$. 2021, ApJ, 917, 79. [2105.11572](#).

Research Experience

Undergraduate Research Course

2021 Sep - Present

Department of Astronomy and Astrophysics, University of Toronto

Supervisor: [Prof. Roberto Abraham](#)

Software for the Dragonfly Telephoto Array.

Independent Research

2021 Sep - Present

Supervisors: [Prof. Gwendolyn Eadie](#), [Prof. Alex Stringer](#)

Diagnostics for non-iterative Bayesian approximations.

Independent Research

2021 Jul - Present

Supervisor: [Prof. Norman Murray](#)

Using Stan to fit dynamical models for the angular momentum of the Earth-Moon system to constrain the strength of thermal tides over geological times.

Independent Research

2021 May - Present

Supervisor: [Dr. Joshua Speagle](#)

Probabilistic predictions of stellar ages with normalizing flows. Quantifying the contribution of stellar and Galactic information to age estimates.

Undergraduate Summer Research Fellowship

2021 May - 2021 Aug

Canadian Institute for Theoretical Astrophysics

Supervisors: [Prof. Gwendolyn Eadie](#), [Prof. Norman Murray](#), [Dr. Joshua Speagle](#)

Continued work on the mass distribution of the Milky Way. Collaborated with members of the [H3 Survey](#) and submitted a first-author paper to ApJ.

Undergraduate Research Course

2020 Aug - 2021 May

Canadian Institute for Theoretical Astrophysics

Supervisors: [Prof. Gwendolyn Eadie](#), [Prof. Norman Murray](#)

Developed a hierarchical Bayesian model for modelling the mass of galactic halos using a fast and scalable sampling algorithm.

Summer Undergraduate Research Program

2020 May - 2020 Aug

Dunlap Institute for Astronomy and Astrophysics, University of Toronto

Supervisor: [Prof. Allison Man](#)

Analyzed ALMA data to infer molecular gas masses in several gravitationally lensed high-redshift galaxies. Presented results in a first-author publication in ApJ.

Presentations

SDSS 2021 Collaboration Meeting, JHU/Online	2021 August
Lightning talk: “Predicting the ages of stars with machine learning.”	
Stellar Stats Workshop, UofT/Online	2021 May
Lightning talk: “Estimating the mass distribution of the Milky Way with Bayesian multilevel models.”	
Canadian Astronomical Society (CASCA) AGM, NRC Herzberg/Online	2021 May
Poster presentation and lightning talk: “Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$.”	
Vancouver Area Cosmology Meeting, UBC/SFU/TRIUMF/Online	2021 Apr
Talk: “Molecular gas in gravitationally lensed galaxies at $z = 2.9$.”	
REQUIEM Galaxies Telecon, Online	2021 Oct
Talk: “Molecular gas in gravitationally lensed galaxies at $z = 2.9$.”	
Summer Undergraduate Research Program Poster Session, UofT/Online	2021 Aug
Poster presentation: “Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$.” Awarded prize for one of top three posters.	

Honours and Awards

Undergraduate Summer Research Fellowship, (\$8,396 CAD)	2021 May - 2021 Aug
Summer Undergraduate Research Program Award (\$9,500 CAD)	2020 May - 2020 Aug
Dunlap Institute for Astronomy and Astrophysics, University of Toronto	

Selected Projects

1. Comprehensive scientific computing package for Rust: linear algebra, statistics, numerical optimization, linear models, and more. github.com/al-jshen/compute
2. Zero-dependency reverse-mode automatic differentiation package. github.com/al-jshen/reverse
3. Fast and easy random number generation in Rust with Wyrand. github.com/al-jshen/alea
4. Parallelized ray tracer for realistic scene rendering. github.com/al-jshen/traycer

Media

National Radio Astronomy Organization (NRAO) eNews	2021 Jul
Digital newsletter feature: “Molecular gas in high redshift galaxies”. https://science.nrao.edu/enews/14.7/index.shtml#molecular_gas	
SURP Student of the Week Feature	2020 Jul
Interview: https://www.dunlap.utoronto.ca/surp-sow/sow12020/	

Outreach

Space Science Journalist (Current in Space segment), <i>The Star Spot</i> podcast	2019 - 2021
Curate, write and record stories about research and news in astronomy. Episodes 177 - end of show.	

Volunteering

Python Workshop Lead, Sir Winston Churchill Secondary, Vancouver, B.C.	2015 - 2018
Coordinated and delivered workshops to prepare students for national programming contest.	

Technical Skills

Data: Stan, Pyro, Pytorch, NumPy, SciPy, Pandas, Scikit-Learn, Matplotlib, SQL, R
Other: C, C++, Rust, BLAS/LAPACK, Astropy, L ^A T _E X, regex, bash, git, Docker, AWS, Javascript, React